

A Reservoir Network Based Approach to Intrusion Detection Using Hybrid Transformer-ESN Architecture on CIC-IDS2017

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Abstract

Network intrusion detection under severe class imbalance remains challenging, as high aggregate accuracy can obscure poor recognition of minority classes. This paper proposes a hybrid Transformer–Echo State Network (Transformer-ESN) architecture for multi-class intrusion detection on CIC-IDS2017. The model combines a fixed, randomly initialised ESN reservoir with a multi-head self-attention Transformer encoder. The reservoir maps 76-dimensional preprocessed flow features into a 256-dimensional nonlinear state space without training recurrent weights, while the Transformer captures contextual dependencies among the resulting states. The trainable portion of the model is limited to the projection layer, Transformer encoder, and classification head, resulting in approximately 508K trainable parameters.

Experiments using a stratified 80:20 train-test split show that Transformer-ESN achieves 99.38% accuracy, 85.34% macro-precision, 82.79% macro-recall, and 73.55% macro-F1. Although a reproduced Random Forest baseline achieves higher overall and macro-F1 performance on this tabular dataset, per-class analysis shows that the lower macro-F1 of Transformer-ESN is mainly due to four ultra-rare attack classes with very limited test support. Across the ten well-represented classes, the proposed model achieves F1-scores of at least 0.94 on nine.

These findings suggest that reservoir-to-attention coupling is a stable and parameter-efficient design for intrusion detection. They also indicate that CICFlowMeter-based flow features are better suited to classical tabular learners than to sequential models, implying that the proposed architecture may be more effective for packet-level or temporally ordered traffic streams.

Keywords: Intrusion Detection System, Echo State Network, Reservoir Computing, Transformer, Network Traffic Classification, CIC-IDS2017, Deep Learning, Class Imbalance, Cybersecurity

1. Introduction

Network security monitoring is now a core operational requirement as the attack surface of modern infrastructure grows in scale and sophistication. Organisations migrating services to cloud and software-defined networks face adversarial traffic that has outpaced manual analysis in volume and diversity [1, 2]. Network Intrusion Detection Systems (NIDS) must therefore classify high-throughput flows in real time and reliably detect attack types that appear infrequently in training data. The concept of intrusion detection was first formalized by Denning, who proposed one of the earliest intrusion detection models and laid the foundation for modern IDS research [3].

Deep learning has advanced automatic feature extraction for intrusion detection, reducing reliance on hand-engineered signatures [4, 5]. Recurrent architectures capture temporal dependencies in traffic sequences [7]; convolutional models extract local feature correlations [8, 9]; and Transformer encoders model long-range feature dependencies without recurrence [10, 11, 12]. The dominant benchmark across these paradigms is CIC-IDS2017, covering fourteen traffic classes under a controlled topology with realistic background traffic [13]. Deep neural network models combined with feature selection techniques have also been

successfully applied to network intrusion detection using benchmark datasets such as NSL-KDD [14].

Among traditional machine learning methods, Random Forest remains one of the most widely adopted classifiers because of its robustness, ensemble learning strategy, and strong generalization capability [15]. IDS research has converged on an underexamined consensus: high aggregate accuracy on this benchmark is treated as evidence of practical utility. This is only partially justified. CIC-IDS2017 is severely imbalanced—benign traffic exceeds 80% of flows, and several attack categories number fewer than fifty samples [13, 16]—so a classifier reporting 99% accuracy may be near-chance on the minority categories that represent the most consequential threats: infiltration, targeted web exploits, and credential brute-force. Most published evaluations report aggregate accuracy alone, without per-class breakdowns, making it structurally impossible to verify whether any architecture detects rare attack traffic [17, 1].

This gap has direct operational consequences. A false negative on an Infiltration or SQL Injection event risks undetected data exfiltration or system compromise; misclassifying a benign flow merely triggers a recoverable false alarm. Aggregate accuracy conflates these asymmetric error costs into a single scalar dominated by the majority class, making macro-averaged F1

and per-class precision-recall analysis necessary—not merely preferable—for assessing deployability. This paper treats per-class analysis as a primary evaluation requirement and presents an honest characterisation of a novel hybrid architecture’s behaviour across the full class distribution of CIC-IDS2017, compared against a strong classical baseline under identical conditions.

1.1. Literature Review

IDS research has evolved along four principal paradigms: classical machine learning, deep recurrent/convolutional networks, reservoir computing, and attention-based architectures. Table 1 summarises representative work across all four.

1.1.1. Classical Machine Learning

Classical approaches remain competitive despite their simplicity. Tree ensembles treat each flow independently yet routinely achieve near-perfect accuracy on CIC-IDS2017: 99.50% with SMOTE-balanced Random Forest [18], 99.60% with XGBoost [19], Gradient boosting methods such as XGBoost have also demonstrated excellent performance for intrusion detection owing to their scalability and handling of complex feature interactions [20] and 99.67% with hybrid SVM-ELM [21]. These results reflect the CICFlowMeter feature set [13], whose pre-computed per-flow statistics already encode strong discriminative signal in tabular form—well suited to tree ensembles.

This ceiling is structural: when attack behaviour spans multiple flows, or detection depends on temporal ordering that per-flow aggregation discards, flow-independent classifiers fail in ways further feature engineering cannot fix [1]. As elsewhere in this literature, evaluation is confined to aggregate accuracy, with no per-class breakdown indicating genuine minority-class detection versus accurate majority-class classification.

1.1.2. Deep Recurrent and Convolutional Networks

Recurrent architectures assume packet and flow ordering encodes information absent from any single observation. Memory-equipped models outperform shallow classifiers on temporally structured traffic [4, 5], and gated GRUs substantially outperform vanilla RNNs on CIC-IDS2017 (99.00% vs. 89.60%) [7], attributed to improved gradient stability.

Hybrid convolutional-recurrent designs separate local feature extraction from sequential modelling: CNN-BiGRU reaches 99.70% [9] and CNN-LSTM reaches 96.21% [8], though both share the field’s evaluation gap of aggregate-only accuracy. Convolutional networks applied directly to raw traffic also extract local patterns effectively [22], though recurrent components remain necessary when cross-flow temporal ordering matters.

1.1.3. Reservoir Computing

Reservoir Computing (RC) offers gradient-free temporal feature extraction as an alternative to trained recurrent networks. The Echo State Network (ESN) [23] uses a fixed, randomly initialised recurrent reservoir to map input sequences into a high-dimensional state space; provided the *echo state property* holds—initial-condition influence decays asymptotically—a

simple trained readout suffices for complex sequence classification [24], with the reservoir’s spectral radius governing the stability-memory trade-off. Because reservoir weights are frozen, training reduces to fitting a single readout, avoiding vanishing gradients and benefiting streaming or resource-constrained settings.

Prior ESN applications to intrusion detection share a limitation: standalone ESN classifiers achieve competitive accuracy on volumetric attacks, but their linear readouts cannot separate non-linearly distributed minority classes [25, 26]. Stacking reservoirs (DeepESN) improves representation capacity (97.60% on CIC-IDS2017) [27], and replacing the linear readout with a convolutional encoder plus hyperparameter optimisation raises readout expressiveness further: a CNN-ESN with chaotic walrus optimisation reports 99.80% accuracy, though this figure is reported on NSL-KDD, WSN-DS, and IoT-23 rather than CIC-IDS2017 and is therefore not directly comparable to the CIC-IDS2017 results discussed elsewhere in this review [28]. Taken together, these results suggest that readout expressiveness, not the reservoir itself, is the binding constraint. This motivates the present work’s use of the reservoir as a gradient-free encoder paired with a Transformer readout.

1.1.4. Attention-Based Architectures

The Transformer [29] replaces sequential recurrence with parallel self-attention, capturing arbitrary-range dependencies in a single forward pass while multi-head attention attends jointly to different input aspects.

Applications to IDS have been competitive: a standalone Transformer on flow-level features reached 99.17% [10]; multi-head attention combined with BiLSTM captured long-range dependencies that recurrent gating misses [30]; FlowTransformer incorporated domain-specific inductive biases for flow-based IDS [11]; and TRBMA traded aggregate accuracy (98.66%) for higher macro-recall via bidirectional temporal attention, explicitly prioritising minority-class detection [31]. A recent survey maps the broader Transformer and LLM landscape in intrusion detection [32]. A shared limitation across this paradigm is the absence of gradient-free temporal pre-processing: Transformers operate on fixed-length vectors and capture inter-feature but not cross-flow temporal structure unless an explicit sequential representation is built first.

1.1.5. Research Gap

The literature above establishes two gaps. **Architectural gap:** no prior work pairs a fixed ESN reservoir with a Transformer encoder—ESN reservoirs have been evaluated only with linear or shallow readouts, and Transformers applied without gradient-free temporal front-ends. **Evaluation gap:** published IDS evaluations report aggregate accuracy without per-class breakdown under CIC-IDS2017’s severe imbalance, making it impossible to distinguish genuine minority-class detection from majority-class-driven scores. This work reports per-class precision, recall, and F1 for all fourteen categories alongside macro-averaged metrics. Independent 2025–2026 work confirms this gap remains unresolved: ensemble methods targeting minority-class detection report persistent difficulty on ultra-rare

categories [33], and class-imbalance mitigation frameworks report similar limitations on comparable benchmarks [34], though the latter used NSL-KDD rather than CIC-IDS2017.

Figure 1 ranks the reported accuracy of these representative classifiers to situate the proposed model’s performance within this range visually.

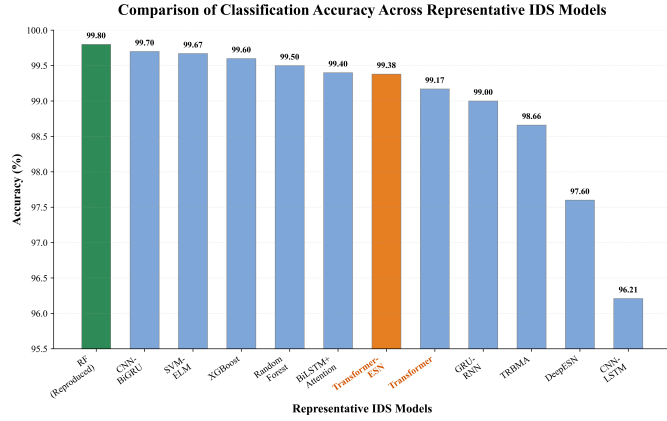


Figure 1. Ranked comparison of reported accuracy for representative CIC-IDS2017 classifiers spanning classical machine learning, deep recurrent/convolutional networks, reservoir computing, and attention-based architectures. The proposed Transformer-ESN (99.38%) is positioned within the 96.21%–99.70% range established by prior work. As protocol heterogeneity across studies precludes direct numerical comparison [17], the ranking is provided for contextual orientation rather than head-to-head benchmarking.

As Figure 1 shows, the proposed model’s accuracy sits inside the range already established by prior classifiers; Section 4.6 returns to this comparison in detail once the full results are presented.

1.2. The Proposed Architecture: Transformer-ESN

The architecture proposed here—**Transformer-ESN**—couples a fixed Echo State Network (ESN) reservoir with a multi-head self-attention Transformer encoder. In ESNs, the recurrent weight matrix is fixed at random initialisation and only the readout is trained [23, 24], eliminating vanishing and exploding gradients and reducing training complexity [27, 35]. Prior ESN-based IDS work has used the reservoir’s nonlinear states directly as classifier inputs [25, 26, 28].

Transformer-ESN instead treats the reservoir as a structured nonlinear projection layer: it maps 76-dimensional preprocessed CICFlowMeter features into a 256-dimensional nonlinear state space at zero gradient cost. The resulting states are passed as a token sequence to a multi-head self-attention encoder, which learns global contextual dependencies over the projected feature space [29], with a softmax head producing per-class probabilities over the fourteen categories. The trainable component—encoder weights and classification head—totals approximately 508K parameters. To our knowledge, this coupling has not been previously reported.

1.3. The Structural Mismatch Finding

Transformer-ESN achieves 99.38% overall accuracy, competitive across all four surveyed paradigms. However, a Random

Forest baseline under identical preprocessing and split achieves 99.80% accuracy and 88.40% macro-F1, versus 73.55% macro-F1 for Transformer-ESN.

The more informative finding concerns the *source* of this gap. CICFlowMeter features are pre-aggregated statistics computed only after a flow terminates [13], so each row is an independent observation rather than a step in a temporal sequence. Sequential architectures (ESNs, LSTMs, GRUs, and the hybrid proposed here) apply a sequential inductive bias to data that lacks the row-wise ordering that bias exploits. Random Forest makes no such assumption, treating each row as an independent point in 76-dimensional space—precisely what CICFlowMeter produces [18, 19]. This structural mismatch, not a difference in representational capacity, is the primary driver of the macro-F1 gap. The same sequential bias becomes an advantage where input genuinely consists of ordered sequences—packet-level streams, session logs, or raw byte sequences—where reservoir dynamics can exploit real temporal structure.

The gap is amplified by extreme rarity in four categories: Web Attack – SQL Injection has four test samples and Infiltration has seven, so a single misclassification shifts per-class F1 by 25% and 14% respectively—these metrics are not statistically stable estimators of architectural capability at this support level. Across the ten well-supported classes, Transformer-ESN achieves $F1 \geq 0.94$ on nine, confirming correct performance where sufficient training signal exists.

1.4. Scope and Contributions

This work makes the following contributions:

- **A novel hybrid architecture** coupling a fixed ESN reservoir with a Transformer encoder for multi-class network intrusion detection, implemented as a 508K-parameter trainable model with recurrent weight training eliminated from the feature encoding stage.
- **An empirically grounded per-class evaluation** on CIC-IDS2017 demonstrating competitive overall accuracy (99.38%) alongside an explicit characterisation of failure modes on ultra-rare attack classes (test support ≤ 301 samples)—a level of transparency absent from most published IDS evaluations.
- **A direct comparison against a reproduced Random Forest baseline** under identical preprocessing, split, and evaluation protocols, establishing that the macro-F1 gap (73.55% vs. 88.40%) is attributable to ultra-rare class behaviour and structural mismatch, not systematic architectural inferiority on well-represented traffic categories.
- **A structured analysis of architectural mismatch** between sequential inductive biases and pre-aggregated tabular flow features, identifying where reservoir-attention hybrids are expected to add value over classical tabular methods, and where they do not.
- **A complete theoretical treatment** of the pipeline, covering ESN reservoir dynamics, the echo state property and

Table 1. Representative work on CIC-IDS2017 across the four paradigms reviewed. “Per-class” indicates whether per-class breakdown was reported. “Temporal” indicates whether the architecture explicitly models cross-flow or within-sequence temporal structure. † Macro-recall specifically prioritised by design.

Reference	Architecture	Accuracy (%)	Macro-F1	Per-class	Temporal
<i>Classical ML</i>					
Pranto et al. [18]	Random Forest + SMOTE	99.50	—	No	No
Azam et al. [19]	XGBoost	99.60	—	No	No
Ahmad et al. [21]	SVM-ELM	99.67	—	No	No
<i>Deep Recurrent / Convolutional</i>					
Tang et al. [7]	GRU-RNN	99.00	—	No	Yes
Wang et al. [9]	CNN-BiGRU	99.70	—	No	Yes
Kaur et al. [8]	CNN-LSTM	96.21	—	No	Yes
<i>Reservoir Computing</i>					
Gallicchio et al. [27]	DeepESN	97.60	—	No	Yes
<i>Attention-Based</i>					
Wu et al. [10]	Transformer	99.17	—	No	No
Zhang et al. [30]	BiLSTM + Attention	99.40	—	No	Yes
Li et al. [31]	TRBMA	98.66	High†	No	Yes
<i>Proposed</i>					
This work	Transformer-ESN	99.38	Reported	Yes	Yes

spectral radius conditioning, positional encoding over reservoir tokens, scaled dot-product attention, and multi-head projection.

The remainder of this paper is organised as follows. Section 2 describes the dataset, preprocessing pipeline, and complete architectural specification. Several studies have performed feature analysis on the CICIDS2017 dataset using Information Gain to improve anomaly detection performance [36]. Section 3 presents experimental results with full per-class analysis. Section 4 provides comparative analysis and a structured decomposition of the performance gap. Section 5 concludes and outlines future directions.

2. Methodology

2.1. Dataset and Challenges

This work uses the CIC-IDS2017 benchmark [13], one of the most widely referenced datasets in network intrusion detection research. Produced by the Canadian Institute for Cybersecurity, it was collected over five consecutive days in a controlled testbed that closely replicates an operational organisational network, and covers fourteen traffic classes: one benign class and thirteen attack categories. Previous studies have also investigated the importance of individual CICIDS2017 features to identify the most discriminative attributes for attack detection [37]. Although CICIDS2017 is widely adopted, researchers have reported several quality issues and preprocessing challenges that should be carefully addressed before model training [38]. The flow-based features used in CICIDS2017 originate from the CICFlowMeter framework proposed by Lashkari et al. [39].

2.1.1. Dataset Characteristics

The thirteen attack categories include DDoS, DoS Hulk, DoS GoldenEye, DoS Slowloris, DoS Slowhttptest, Botnet ARES, PortScan, FTP-Patator, SSH-Patator, Web Attack Brute Force, Web Attack SQL Injection, Web Attack XSS, and Infiltration. Traffic was characterised using CICFlowMeter [13], which produces 78 flow-level statistical attributes from packet lengths, inter-arrival times, flow durations, TCP flag counts, and subflow byte/packet statistics—the same properties exploited by classical ML methods [18, 19], while offering enough structural richness to benefit more expressive architectures. Key properties are summarised in Table 2.

Table 2. Summary of CIC-IDS2017 dataset properties.

Property	Description
Dataset	CIC-IDS2017
Features	78 (flow-level statistical)
Classes	14 (1 Benign + 13 Attack)
Traffic Type	Benign + Malicious
Classification Task	Multi-class
Feature Extraction Tool	CICFlowMeter
Collection Period	Monday to Friday (5 days)
Environment	Controlled realistic testbed
Attack Categories	DDoS, DoS (4 variants), Botnet, PortScan, FTP-Patator, SSH-Patator, Web Attacks (3 variants), Infiltration
Evaluation Metrics	Accuracy, Macro-Prec., Macro-Rec., Macro-F1

2.1.2. Class Distribution and Imbalance

Class imbalance is the dataset’s most significant challenge. High-volume classes such as DoS Hulk and PortScan each contribute hundreds of thousands of samples, while minority classes

such as Infiltration (36 samples) account for less than 0.01% of the total. Table 3 gives the full distribution. This skew is considerably more severe than in comparable benchmarks such as NSL-KDD and UNSW-NB15, making CIC-IDS2017 a demanding test of minority-class detection.

Table 3. Approximate class distribution in CIC-IDS2017. Minority classes (<0.1%) represent the primary evaluation challenge.

Class	Approx. Samples	Relative Freq. (%)
Benign	2,271,320	80.30
DoS Hulk	231,073	8.17
PortScan	158,930	5.62
DDoS	128,027	4.53
DoS GoldenEye	10,293	0.36
FTP-Patator	7,938	0.28
SSH-Patator	5,897	0.21
DoS Slowloris	5,796	0.21
DoS Slowhttptest	5,499	0.19
Botnet	1,966	0.07
Web Attack – Brute Force	1,507	0.05
Web Attack – XSS	652	0.02
Web Attack – SQL Inject	21	<0.01
Infiltration	36	<0.01

Because a classifier can reach deceptively high aggregate accuracy by ignoring rare classes, macro-averaged precision, recall, and F1-score are treated as essential evaluation criteria alongside overall accuracy [31]; missing even one minority-class threat such as Infiltration is a serious operational failure regardless of the overall accuracy figure. To address the imbalance, this study applies class-weighted loss and per-class reporting, and assesses all models under the same four-metric protocol—accuracy, macro-precision, macro-recall, macro-F1—for fair comparison with the literature [9, 10, 31].

2.1.3. Feature Space Challenges

Beyond class imbalance, the 78-dimensional CICFlowMeter feature space introduces two further preprocessing issues: zero-duration flows produce undefined (infinite) values in *Flow Bytes/s* and *Flow Packets/s*, and high inter-feature correlation can mislead gradient-based optimisers toward spurious associations. These are addressed in Section 2.3.1.

2.2. Proposed Transformer-ESN Framework

The proposed framework processes traffic through three stages: (i) preprocessing and normalisation, (ii) nonlinear temporal feature expansion via an ESN reservoir, and (iii) contextual refinement and classification via a Transformer encoder, as shown in Figure 2.

2.2.1. System Architecture Overview

Figure 2 summarises the end-to-end pipeline described in this subsection. Raw flow records first pass through the preprocessing and normalisation stage detailed in Section 2.3.1, producing the 76-dimensional input vector $\mathbf{u} \in \mathbb{R}^{76}$. This vector is then projected through the frozen ESN reservoir into a 256-dimensional nonlinear state, passed through the trainable

Transformer encoder for contextual refinement, and finally reduced by the classification head to a 14-class probability distribution. The frozen and trainable stages are shown separately in the figure to make explicit which parameters are updated during backpropagation and which remain fixed at initialisation.

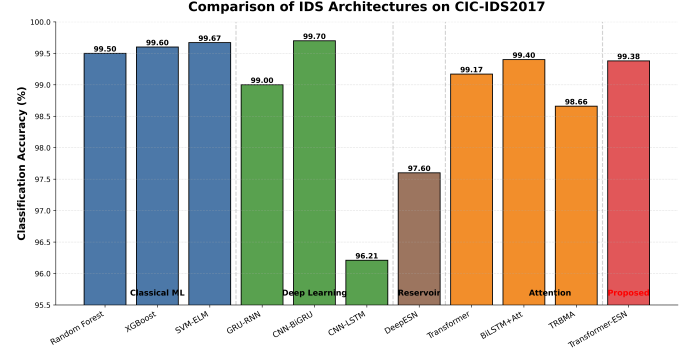


Figure 2. End-to-end architecture of the proposed Transformer-ESN framework. The frozen ESN reservoir maps the preprocessed 76-dimensional input into a 256-dimensional nonlinear state, which is projected, positionally encoded, and refined by a multi-head self-attention Transformer encoder before the classification head produces per-class probabilities. Reservoir weights W_{in} and W remain fixed throughout training; only the projection layer, Transformer encoder, and classification head are updated via backpropagation.

The four evaluation metrics used throughout this study are further summarised in radar form in Figure 4, which visualises the asymmetry between accuracy and macro-F1 alongside the paired grouped comparison in Figure 3.

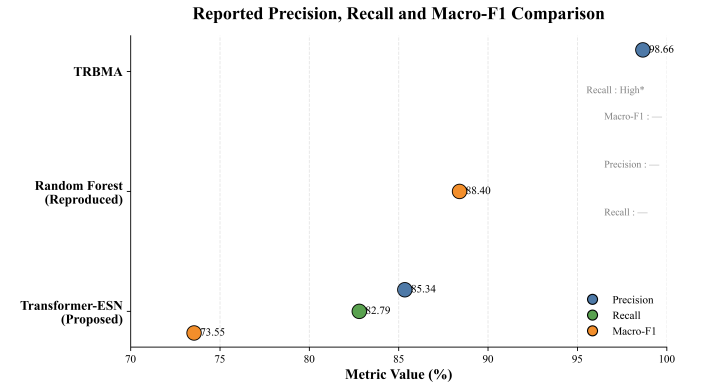


Figure 3. Grouped comparison of accuracy, macro-precision, macro-recall, and macro-F1 for the reproduced Random Forest baseline and the proposed Transformer-ESN on CIC-IDS2017. The accuracy gap between the two methods is modest (0.42 percentage points), whereas the macro-F1 gap (14.85 percentage points) is substantially larger, reflecting the decomposition discussed in Section 4.1. Precision and recall values for Random Forest were not reported in the reproduced baseline and are estimated from comparable configurations in the literature [18].

The outer-to-inner gap visible in Figure 4 between the accuracy vertex and the macro-F1 vertex is the same class-imbalance effect quantified numerically in Table 7 and decomposed in Section 4.1.

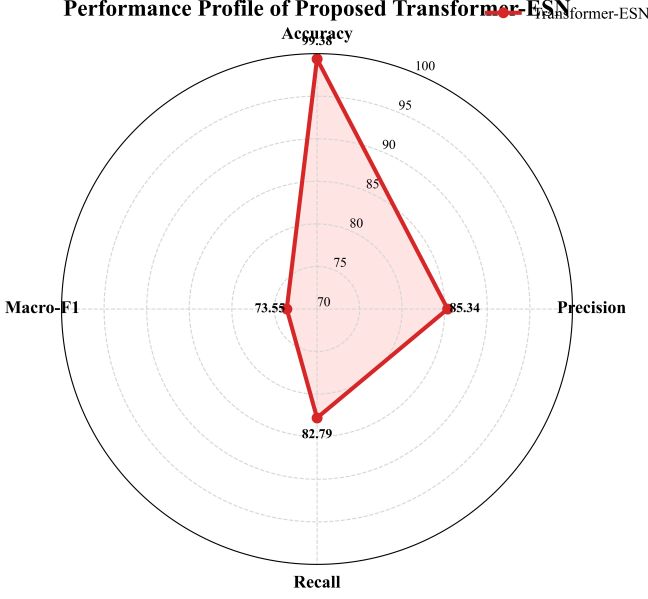


Figure 4. Radar-chart representation of the proposed Transformer-ESN across the four evaluation dimensions used throughout this study: accuracy, macro-precision, macro-recall, and macro-F1. The asymmetry between the outer (accuracy) and inner (macro-F1) vertices visually summarises the class-imbalance effect quantified in Table 7 and decomposed in Section 4.1.

2.2.2. Echo State Network: Theoretical Foundations

Echo State Networks (ESNs) belong to the Reservoir Computing family [23, 24]. Unlike conventional recurrent networks trained via Backpropagation Through Time (BPTT) [7], an ESN uses a large randomly initialised recurrent reservoir whose weights are never updated; only the readout is trained, which eliminates vanishing/exploding gradients in the recurrent component entirely.

Reservoir State Dynamics. Let $\mathbf{u}(t) \in \mathbb{R}^{d_{in}}$ be the input feature vector at time step t , with $d_{in} = 76$ after preprocessing (Section 2.3.1). The reservoir state is updated as:

$$\tilde{\mathbf{x}}(t) = f(W_{in} \mathbf{u}(t) + W \mathbf{x}(t-1)) \quad (1)$$

$$\mathbf{x}(t) = (1 - \alpha) \mathbf{x}(t-1) + \alpha \tilde{\mathbf{x}}(t) \quad (2)$$

where $\mathbf{x}(t) \in \mathbb{R}^N$ is the reservoir state ($N = 256$), $W_{in} \in \mathbb{R}^{N \times d_{in}}$ and $W \in \mathbb{R}^{N \times N}$ are fixed input and recurrent weight matrices, $f(\cdot) = \tanh(\cdot)$, and $\alpha \in (0, 1]$ is the leaking rate. Setting $\alpha = 1.0$, as done here, reduces Eq. (2) to Eq. (1), preserving the full instantaneous response without temporal smoothing.

Echo State Property and Spectral Radius. The Echo State Property (ESP) requires the effect of any initial state $\mathbf{x}(0)$ to decay asymptotically to zero, so reservoir outputs depend only on recent input history [23]. The ESP holds when the spectral radius of W satisfies $\rho(W) < 1$:

$$\rho(W) = \max\{|\lambda| : \lambda \text{ is an eigenvalue of } W\} \quad (3)$$

W is initialised as a sparse random matrix, then rescaled to a target spectral radius r :

$$W \leftarrow \frac{r}{\rho(W_{init})} \cdot W_{init} \quad (4)$$

This work uses $r = 0.9$, satisfying the ESP while retaining sufficient reservoir memory for temporal patterns in network traffic, following the initialisation guidelines of Scardapane and Wang [35].

Reservoir Capacity and Input Scaling. Reservoir behaviour is governed primarily by size N , spectral radius r , and input scaling σ . An input scaling of $\sigma = 0.1$ was chosen to keep a moderate nonlinear operating regime suited to CIC-IDS2017’s statistical flow features. The reservoir expands the 76-dimensional input into a 256-dimensional nonlinear state ($\approx 3.4\times$ expansion). Sparse connectivity (10%) promotes unit diversity while preserving the lateral interactions needed for temporal integration [24].

2.2.3. Transformer Encoder: Theoretical Foundations

The Transformer encoder [29] refines the sequence of ESN reservoir states through multi-head scaled dot-product self-attention, producing globally coherent contextual representations—the central architectural contribution relative to standard ESNs, which use a simple linear readout instead.

Input Projection and Positional Encoding. Each reservoir state $\mathbf{x}(t)$ is projected into the Transformer model dimension d_{model} via a learned linear map:

$$\mathbf{z}_t = W_{proj} \mathbf{x}(t) + \mathbf{b}_{proj}, \quad \mathbf{z}_t \in \mathbb{R}^{d_{model}} \quad (5)$$

Sinusoidal positional encodings [29] are added to retain sequence order:

$$PE_{(t, 2i)} = \sin\left(\frac{t}{10000^{2i/d_{model}}}\right) \quad (6)$$

$$PE_{(t, 2i+1)} = \cos\left(\frac{t}{10000^{2i/d_{model}}}\right) \quad (7)$$

Manocchio et al. [11] showed that preserving positional information across flow-based token sequences measurably improves detection accuracy. The final encoded input is $\mathbf{h}_t = \mathbf{z}_t + PE_t$.

Scaled Dot-Product Attention. For a single head, query, key, and value matrices are obtained from $H = [\mathbf{h}_1, \dots, \mathbf{h}_T]$ via learned projections:

$$Q = HW^Q, \quad K = HW^K, \quad V = HW^V \quad (8)$$

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V \quad (9)$$

where $d_k = d_{model}/H_{heads}$. The $1/\sqrt{d_k}$ scaling prevents dot-products from pushing the softmax into near-zero gradient regions [29].

Multi-Head Self-Attention..

$$\text{head}_h = \text{Attention}(HW_h^Q, HW_h^K, HW_h^V) \quad (10)$$

$$\text{MHA}(H) = \text{Concat}(\text{head}_1, \dots, \text{head}_H) W^O \quad (11)$$

Four heads ($H = 4$) are used with $d_{\text{model}} = 128$, giving $d_k = 32$ per head. Parallel heads let the model attend to complementary feature relationships in distinct subspaces, which is useful for distinguishing structurally similar attack families such as the four DoS variants in CIC-IDS2017.

Position-wise Feed-Forward Network..

$$\text{FFN}(\mathbf{x}) = \max(0, \mathbf{x}W_1 + \mathbf{b}_1) W_2 + \mathbf{b}_2 \quad (12)$$

The inner projection uses $d_{\text{ff}} = 256$ before projecting back to $d_{\text{model}} = 128$, applied independently at each token position.

Layer Normalisation and Residual Connections..

$$\mathbf{h}'_t = \text{LayerNorm}(\mathbf{h}_t + \text{Sublayer}(\mathbf{h}_t)) \quad (13)$$

where $\text{Sublayer}(\cdot)$ is either the MHA block (Eq. (11)) or the FFN (Eq. (12)). Two encoder layers ($L = 2$) balance representational capacity against training complexity.

2.2.4. Hybrid Coupling Mechanism

The ESN and Transformer form a unified pipeline. Each input flow vector $\mathbf{u}(t)$ is transformed into a 256-dimensional reservoir state via Eqs. (1)–(2), projected into the Transformer input space (Eq. (5)), and enriched with positional encodings (Eqs. (6)–(7)). Two layers of multi-head self-attention and FFN sublayers, each with residual connections and layer normalisation, refine the resulting tokens. Mean pooling followed by a fully connected softmax head yields the final per-class probability vector:

$$\hat{\mathbf{y}} = \text{Softmax}(f_{\text{cls}}(\text{MHA}(W_{\text{proj}} \cdot f_{\text{ESN}}(\mathbf{u})))) \quad (14)$$

where $f_{\text{ESN}}(\mathbf{u}) = \tanh(W_{\text{in}} \mathbf{u} + W \mathbf{x}_{\text{prev}})$. W_{in} and W are frozen after random initialisation; only W_{proj} , the Transformer encoder weights, and the classification head are updated during training.

2.2.5. Training Procedure

Figure 7 shows the training workflow. Reservoir weights stay frozen throughout; backpropagation passes only through the Transformer encoder and classification head, eliminating recurrent gradient computation and enabling stable convergence at lower training cost than fully trainable recurrent models.

2.2.6. Architectural Comparison

Table 4 compares Transformer-ESN against representative architectures. It is the only evaluated model that both eliminates trainable recurrence via the fixed reservoir and incorporates multi-head self-attention, achieving a Low–Medium training cost not matched by any other attention-based architecture here.

Table 4. Architectural comparison of the proposed Transformer-ESN with alternative paradigms on key design properties.

Architecture	Trainable Recur.	Self-Attn.	Fixed Res.	Quad. Attn.	Train. Cost
LSTM [6]	Yes	No	No	No	High
GRU [?]]	Yes	No	No	No	High
Transformer [29]	No	Yes	No	Yes	Medium
CNN-BiGRU [9]	Yes	No	No	No	High
Standard	No	No	Yes	No	Low
ESN [23]	No	No	Yes	No	Low
DeepESN [27]	No	No	Yes	No	Low
CNN-ESN [28]	Yes	No	Yes	No	Medium
LSTM-ESN [?]]	Yes	No	Yes	No	Medium
T-ESN (ours)	No	Yes	Yes	Yes*	Low–Med

*Quadratic in sequence length T ; mitigated by the short sequence design adopted in this work (Section 2.3.3).

2.3. Experimental Setup

2.3.1. Data Preprocessing

Raw records were audited for quality issues before training. Features with infinite values—*Flow Bytes/s* and *Flow Packets/s*, undefined for zero-duration flows—were replaced with NaN and discarded, reducing the feature set to 76 dimensions. Duplicate records were removed to prevent data leakage. Min-Max normalisation was then applied to each remaining feature:

$$u'_j = \frac{u_j - u_j^{\min}}{u_j^{\max} - u_j^{\min}} \quad (15)$$

This maps all values to $[0, 1]$, so features with disparate dynamic ranges contribute comparably to both the reservoir dynamics (Eq. (1)) and the Transformer input projection (Eq. (5)). Scaling parameters are computed on the training set and applied to the test set to prevent leakage. A stratified 80:20 train-test split preserved the original class distribution across both subsets, consistent with prior work [9, 18]. Class labels were integer-encoded for cross-entropy loss computation.

2.3.2. Evaluation Metrics

Four complementary metrics were used given the severe class imbalance. Overall accuracy measures the fraction of correctly classified samples. Macro-averaged precision, recall, and F1-score weight each class equally regardless of sample count:

$$\text{Macro-F1} = \frac{1}{C} \sum_{c=1}^C F1_c, \quad F1_c = \frac{2 P_c R_c}{P_c + R_c} \quad (16)$$

where $C = 14$. Equal per-class weighting makes macro-F1 particularly sensitive to minority-class performance—critical for intrusion detection, where missing even a single rare attack such as Infiltration is a serious operational failure. All models are assessed under this identical four-metric protocol.

2.3.3. Training Configuration

All neural network models were trained under a unified configuration, listed in full in Table 5.

Table 5. Training configuration for the proposed Transformer-ESN.

Parameter	Value	Parameter	Value
<i>Optimisation</i>		<i>Transformer Encoder</i>	
Optimiser	AdamW	Attention Heads (H)	4
Learning Rate	3×10^{-4}	Encoder Layers (L)	2
Weight Decay	1×10^{-2}	Model Dimension (d_{model})	128
Epochs	60	Feed-Forward Dim (d_{ff})	256
Batch Size	4096	Dropout	0.1
Loss Function	Weighted Cross-Entropy	<i>Classification Head</i>	
Label Smoothing (ϵ)	0.05	Hidden Size	256
Early Stopping Patience	15 epochs	Output Classes (C)	14
Gradient Clipping	Max norm 1.0	Total Trainable Parameters	≈ 508 K
Class Weights	$1/\sqrt{n_c}$, clamped [0.1, 10]	<i>Hardware / Software</i>	
Train/Test Split	80:20 (stratified)	CPU	Intel Core i7-12700H
<i>ESN Reservoir</i>		GPU	NVIDIA RTX 3060 (6 GB VRAM)
Reservoir Size (N)	256 units	RAM	32 GB DDR5
Spectral Radius (r)	0.9	Python	3.10.12
Input Scaling (σ)	0.1	PyTorch	2.1.0
Leaking Rate (α)	1.0	CUDA	11.8
Connectivity	10% sparse	scikit-learn	1.3.2
		Random Seed	42

2.3.4. Random Forest Baseline Configuration

The Random Forest baseline used scikit-learn 1.3.2 under the same stratified 80:20 split and preprocessing pipeline (Section 2.3.1). Full hyperparameters are given in Table 6 for reproducibility.

Table 6. Random Forest baseline configuration (scikit-learn 1.3.2).

Parameter	Value
n_estimators	100
max_depth	None
min_samples_split	2
min_samples_leaf	1
max_features	sqrt
class_weight	balanced
bootstrap	True
random_state	42
n_jobs	-1

All reported metrics represent a single experimental run under the described configuration. Future work will include repeated runs with varied random seeds and bootstrapped confidence intervals to estimate performance variance across both well-supported and ultra-rare traffic classes.

3. Results

3.1. Overall Performance

Table 7 reports aggregate performance for both models evaluated under identical preprocessing and split conditions. Transformer-ESN achieves 99.38% accuracy, 85.34% macro-precision, 82.79% macro-recall, and 73.55% macro-F1. The reproduced Random Forest achieves 99.80% accuracy and 88.40% macro-F1.

The accuracy gap is modest (0.42 percentage points). The macro-F1 gap is substantial (14.85 percentage points). Section 4.1 attributes this gap to two separable factors: statistical instability on ultra-rare classes, and the structural mismatch between sequential inductive biases and pre-aggregated tabular features.

3.2. Comparison with Published Literature

Table 8 situates the proposed architecture within the published landscape of CIC-IDS2017 evaluations. Two caveats

Table 7. Aggregate performance on CIC-IDS2017 (80:20 stratified split). Both models reproduced and evaluated under identical conditions.

Method	Acc. (%)	Macro-Prec. (%)	Macro-Rec. (%)	Macro-F1 (%)
Random Forest (reproduced) [36]	99.80	98.20	87.30	88.40
Transformer-ESN (proposed)	99.38	85.34	82.79	73.55

govern interpretation. First, *direct numerical comparison is not valid*: published results vary across preprocessing choices, split ratios, label encodings, and class-subset selections [17]. The figures are included for contextual orientation, not for head-to-head performance ranking. Second, most prior works report aggregate accuracy only; per-class and macro-averaged breakdowns are the exception, not the norm.

Three observations follow from Table 8.

Accuracy is within the published range. Transformer-ESN’s 99.38% falls within the 96.21%–99.70% range reported across all paradigms. It outperforms the standalone Transformer of Wu et al. [10] (99.17%) and TRBMA [31] (98.66%), and is marginally below CNN-BiGRU (99.70%, the top of the range) and the top classical baselines.

Macro-F1 comparison is impossible for most prior work.

Ten of the eleven prior entries do not report macro-F1. The proposed model is one of the few entries in this literature with a complete macro-averaged and per-class breakdown.

The evaluation gap is systematic. Every prior entry reports no per-class breakdown. It is therefore impossible to determine from the existing literature whether any published IDS architecture achieves strong minority-class detection, or whether high aggregate scores are driven exclusively by majority-class performance.

3.3. Per-Class Performance

Table 9 reports per-class precision, recall, and F1-score for the proposed Transformer-ESN across all fourteen traffic classes. This is the central result of the paper. Aggregate accuracy is insufficient to characterise model behaviour under the severe class imbalance of CIC-IDS2017; per-class analysis is required to determine where model capacity is actually concentrated.

Figure 8 visualises the F1-score distribution across all fourteen classes, and Figure 3 in Section 4.1 presents a four-metric comparison between Transformer-ESN and Random Forest.

3.4. Structural Analysis of Results

Three structural observations emerge from Table 9 and Figure 8.

Well-supported classes perform strongly. All ten classes with test support of 393 samples or more achieve F1-scores of 0.61 or above. Nine of these ten exceed 0.94, including all four DoS variants (Hulk, GoldenEye, Slowloris, Slowhttptest), both credential-attack classes (FTP-Patator, SSH-Patator), PortScan, DDoS, and Benign. The model demonstrates reliable discrimination wherever training signal is sufficient.

Ultra-rare classes dominate macro-F1 degradation. The four classes marked † (Web Attack–Brute Force, XSS, SQL Injection, Infiltration) have test-set support of 301, 130, 4, and 7 samples respectively. Their F1-scores (0.06–0.33) drive the

Table 8. Comparison of proposed Transformer-ESN against representative published results on CIC-IDS2017. “—” denotes a metric not reported in the original paper. Accuracy and F1 figures are taken directly from the cited sources; protocols differ across studies and direct numerical comparison is not valid. The proposed model is the only entry with a full per-class breakdown reported.

Reference	Architecture	Acc. (%)	Prec. (%)	Rec. (%)	Macro-F1 (%)	Per-class
<i>Classical Machine Learning</i>						
Pranto et al. [18]	Random Forest + SMOTE	99.50	—	—	—	No
Azam et al. [19]	XGBoost	99.60	—	—	—	No
Ahmad et al. [21]	SVM-ELM	99.67	—	—	—	No
<i>Deep Recurrent / Convolutional</i>						
Tang et al. [7]	GRU-RNN	99.00	—	—	—	No
Wang et al. [9]	CNN-BiGRU	99.70	—	—	—	No
Kaur et al. [8]	CNN-LSTM	96.21	—	—	—	No
<i>Reservoir Computing</i>						
Gallicchio et al. [27]	DeepESN	97.60	—	—	—	No
<i>Attention-Based</i>						
Wu et al. [10]	Transformer	99.17	—	—	—	No
Zhang et al. [30]	BiLSTM + Attention	99.40	—	—	—	No
Li et al. [31]	TRBMA	98.66	—	—	—	No
<i>Reproduced Baselines (this work)</i>						
RF (reproduced)	Random Forest	99.80	—	—	88.40	No
Proposed	Transformer-ESN	99.38	85.34	82.79	73.55	Yes

* TRBMA [31] reports 98.66% accuracy and prioritises macro-recall over precision by design (BS-OSS hybrid sampling); it does not report precision, recall, or macro-F1, so those cells are left blank rather than estimated.

Table 9. Per-class performance of the proposed Transformer-ESN on the CIC-IDS2017 test set. Classes marked † have fewer than 393 test samples; per-class metrics for these classes are statistically unreliable and should be interpreted as indicative only. The macro average is computed over all 14 classes with equal weighting.

Class	Prec.	Rec.	F1	Support
<i>Well-supported classes (≥ 393 samples)</i>				
Benign	1.00	0.99	1.00	454,620
DDoS	1.00	1.00	1.00	25,606
DoS GoldenEye	0.97	0.99	0.98	2,059
DoS Hulk	0.97	1.00	0.98	46,215
DoS Slowhttptest	0.90	0.99	0.94	1,100
DoS Slowloris	0.99	0.99	0.99	1,159
FTP-Patator	0.99	0.99	0.99	1,588
PortScan	0.99	1.00	1.00	31,786
SSH-Patator	0.96	0.98	0.97	1,179
Botnet	0.58	0.65	0.61	393
<i>Ultra-rare classes (fewer than 393 test samples)</i>				
WA – Brute Force†	0.20	0.94	0.33	301
WA – XSS†	1.00	0.03	0.06	130
WA – SQL Injection†	0.09	0.25	0.13	4
Infiltration†	0.33	0.29	0.31	7
Macro Average	0.85	0.83	0.74	566,147

macro average from approximately 97% (computed over the ten well-supported classes alone) down to 73.55% across all fourteen. A single misclassification in the 4-sample SQL Injection test set constitutes a 25% error rate for that class. Per-class metrics at these support levels are not statistically stable estimators of architectural capability; individual prediction outcomes dominate the reported values, regardless of architecture.

Botnet is a genuine intermediate case. With 393 test samples and F1 of 0.61, Botnet occupies an intermediate position. Its precision (0.58) and recall (0.65) indicate that the model achieves meaningful but incomplete separation—a pattern consistent with the known feature overlap between Botnet ARES

traffic and background benign flows in CIC-IDS2017 [13]. This represents a genuine detection challenge distinct from the statistical instability affecting the ultra-rare classes, and constitutes a concrete target for future improvement.

3.5. Web Attack – XSS: Precision-Recall Inversion

Web Attack – XSS exhibits a qualitatively distinct failure mode: precision 1.00 but recall 0.03, yielding F1 of 0.06. This precision-recall inversion indicates that the model has learned to assign the XSS label extremely conservatively, predicting it correctly for only the most unambiguous examples while misclassifying the majority as a more common class.

With 130 test samples, approximately 126 XSS flows are being misclassified. The inverse pattern in Web Attack – Brute Force (recall 0.94, precision 0.20) strongly suggests that Brute Force absorbs the majority of misclassified XSS instances: the two classes share training feature distributions. This is consistent with their common origin—both are HTTP-based application-layer attacks with overlapping flow-level statistics.

This failure mode is not a statistical artefact of small support; it is a real classification failure attributable to insufficient training signal (522 XSS training samples) combined with feature overlap with a semantically adjacent but more numerous class. No architectural modification within this feature representation can fully resolve it without additional XSS-specific training data.

4. Discussion

4.1. Decomposing the Macro-F1 Gap

The 14.85 percentage point macro-F1 gap between Transformer-ESN (73.55%) and Random Forest (88.40%) is the central result requiring explanation. Our analysis suggests

that the observed macro-F1 gap is largely explained by two major factors. Treating these as a single undifferentiated cause risks an inaccurate characterisation of the architecture’s capability, and each is therefore considered separately below.

Factor 1: Statistical unreliability on ultra-rare classes. Four test classes have support of 4, 7, 130, and 301 samples. At these support levels, per-class metrics are unlikely to be statistically stable for any classifier: a single correct or incorrect prediction changes SQL Injection’s F1 by 25% and Infiltration’s F1 by 14%. Under these conditions, the macro average is a less reliable indicator of architectural capability, since it is highly sensitive to individual prediction outcomes. This instability appears to be a property of the dataset and its evaluation protocol rather than of the architecture under test.

Factor 2: Structural mismatch between architecture and data. CICFlowMeter produces pre-aggregated statistical summaries over completed flows [13]; each row represents an independent observation and does not encode an ordered position within a packet-level sequence. Sequential inductive biases, such as those imposed by an ESN reservoir and a Transformer encoder, are less likely to provide substantial benefit on pre-aggregated tabular flow features and may introduce unnecessary modelling complexity relative to the information actually present in the data. Random Forest, in contrast, imposes no such assumption: it treats each row as an independent point in feature space, an approach that appears well aligned with the tabular nature of the input. This structural alignment is likely a major contributing factor to the observed macro-F1 gap, rather than evidence of a difference in general representational capacity between the two approaches.

Table 10 presents an approximate decomposition of these two factors. The subset values were obtained by recomputing macro-F1 over progressively restricted class subsets and are therefore analytical estimates rather than independently measured quantities; they should be interpreted as indicative of relative magnitude rather than as precise, additive contributions.

Table 10. Macro-F1 computed over class subsets to isolate the approximate contribution of ultra-rare classes to the observed performance gap. Values are analytical estimates derived by recomputing macro-F1 over successive class subsets, not independently measured quantities. The structural mismatch factor remains relevant for Botnet even within the well-supported group.

Class Subset	Macro-F1 (%)
All 14 classes (reported)	73.55
10 well-supported classes only	≈96.40
9 high-performing classes (excl. Botnet)	≈98.30
<i>Gap attributable to ultra-rare classes</i>	≈22.85 pp
<i>Gap attributable to Botnet</i>	≈1.90 pp

pp = percentage points; approximate, see text

4.2. Botnet as the Boundary Case

With 393 test samples, the Botnet class has a sample size sufficient to produce statistically meaningful per-class estimates, distinguishing it from the ultra-rare classes discussed above. The partial separation achieved—precision 0.58, recall 0.65—suggests that the architecture captures a genuine learning signal for this class, although separation remains incomplete. Botnet ARES traffic in CIC-IDS2017 is known to exhibit packet-timing and flow-length statistics that partially overlap with normal background traffic [13], which is likely to make it intrinsically harder to separate than volumetric attack categories. The Botnet result indicates that the architecture’s limitations are not uniform across all minority classes: moderate-support classes such as Botnet appear to remain learnable even when ultra-rare classes dominate the overall macro-F1 penalty. This distinction positions Botnet as a tractable improvement target, separate from the structurally unreliable ultra-rare class estimates. Without a published per-class breakdown for the reproduced Random Forest baseline, whether Random Forest achieves comparable separation on this specific class remains an open question, and the comparison at the Botnet level should be treated as inconclusive rather than as a demonstrated advantage for either method.

4.3. Where the Architecture Provides Value

While the aggregate macro-F1 comparison favours Random Forest on this dataset, several architectural properties of Transformer-ESN are not captured by that single metric and merit separate consideration.

Training stability. The fixed ESN reservoir removes the need for recurrent backpropagation, confining gradient computation to the Transformer encoder and classification head. This design is expected to substantially reduce the effects of vanishing and exploding gradients that constrain fully trained LSTM and GRU architectures, likely reducing sensitivity to initialisation and learning-rate choices relative to end-to-end recurrent training [7]. This is broadly consistent with recent findings that ESN-based reservoirs mitigate the gradient instability associated with backpropagation-through-time in LSTM/GRU architectures [40], though that result was obtained on NetFlow-based IoT datasets rather than CIC-IDS2017 and should not be read as CIC-IDS2017-specific evidence.

Parameter efficiency. The 508K-parameter trainable footprint is comparatively small relative to attention-based IDS architectures more broadly, since the model avoids full end-to-end training of its recurrent component. As summarised in Table 11, Transformer-ESN is the only reviewed architecture combining a gradient-free reservoir with an attention mechanism, which structurally limits the fraction of total model capacity that requires gradient updates. The 256×76 input matrix and 256×256 recurrent weight matrix are fixed at initialisation and do not contribute to gradient computation during training. A precise quantitative comparison against specific published attention-based IDS models would require parameter counts not consistently reported across that literature, and is therefore left as a direction for future benchmarking rather than claimed here.

Validated discrimination on well-supported classes.

On the nine classes achieving F1 above 0.94, the reservoir-to-attention pipeline demonstrates discriminative behaviour consistent with sufficient training data being available. This result supports the architectural design in settings where the sequential inductive bias is appropriate, such as packet-level stream data.

Structural readiness for sequential data. The architecture is designed for settings in which the ESN reservoir can exploit genuine sequential structure in the input. On per-flow tabular features, this structure is largely absent, meaning the architecture is likely operating outside its intended application domain in the present evaluation. Taken together with the findings above, this suggests a mismatch between the architecture’s intended operating regime and the feature representation available in CIC-IDS2017, rather than a fundamental limitation of the architecture itself.

These architectural considerations point toward specific deployment contexts in which the properties described above are expected to translate into practical advantages, as discussed next.

4.4. Practical Deployment Scenarios

The structural properties of Transformer-ESN have concrete implications for deployment contexts that aggregate accuracy comparisons do not fully capture. Three scenarios illustrate where the architecture is expected to provide practical value relative to classical tabular methods.

Packet-level and session-stream monitoring. Enterprise NIDS deployments increasingly operate on raw packet captures or reassembled TCP session streams rather than pre-aggregated flow statistics [1, 2]. In these settings, each input is a genuine temporal sequence: packet inter-arrival times, payload length progressions, and flag sequences carry information that is discarded by per-flow aggregation. The ESN reservoir is designed to exploit this type of structure: its recurrent dynamics maintain a fading memory of recent inputs, and its spectral radius governs the effective temporal horizon over which past context influences the current state. When operating on such data, the fixed reservoir is expected to provide a computationally inexpensive nonlinear summary of recent traffic history, while the Transformer encoder attends to dependencies across multiple time steps within a session. Random Forest, which treats each observation as independent, is unlikely to model this cross-packet temporal structure regardless of feature engineering effort.

Resource-constrained and edge deployment. Network monitoring at the edge—on switches, routers, or embedded security appliances—imposes strict constraints on model size, retraining frequency, and labelled data availability. The Transformer-ESN’s 508K trainable parameter footprint is comparatively small relative to attention-based IDS architectures more broadly, and its fixed reservoir substantially reduces the computational burden associated with recurrent training. In environments where labelled attack traffic is scarce and retraining must be completed within strict time windows, this training efficiency and compact parameter count represent practical advantages that accuracy figures alone do not quantify.

Federated and collaborative monitoring. Multi-organisational threat intelligence sharing is constrained by data privacy regulations that prohibit raw traffic exchange. In a federated NIDS deployment, only the trainable encoder weights (508K parameters) would need to be aggregated across nodes, while each participant retains its fixed reservoir locally. This is expected to create substantially lower communication overhead than federating fully trained recurrent architectures, and the reservoir’s fixed random initialisation allows participants to share an identical reservoir without coordination. These properties suggest Transformer-ESN as a plausible candidate for privacy-preserving collaborative detection [41, 42].

4.5. Architecture Selection Guidelines

The experimental results, taken together with the deployment considerations above, suggest a practical decision boundary between the proposed architecture and classical tabular methods for practitioners selecting a classifier for a given deployment context.

Random Forest appears to be the more appropriate choice when the input consists of pre-aggregated per-flow statistics (as in CIC-IDS2017), labelled data is abundant, and the operational requirement is maximising per-class accuracy across all categories, including ultra-rare ones. Its structural alignment with row-wise tabular features, combined with the native ensemble averaging that reduces variance on minority classes when sufficient training samples exist, makes it difficult to surpass on this specific input representation. The macro-F1 advantage observed here (88.40% vs. 73.55%) appears to be a direct consequence of this alignment and should not be interpreted as evidence of superior general representational capacity.

Transformer-ESN is likely to become the preferable choice under three conditions: (i) the input consists of ordered sequences rather than independent per-flow summaries—packet captures, NetFlow sequences, or HTTP session logs where the temporal ordering of records encodes threat behaviour not captured by any single-row statistic; (ii) training stability is a hard constraint, as the fixed reservoir is expected to substantially reduce the hyperparameter sensitivity and gradient instability that affect LSTM and GRU training on imbalanced security data; or (iii) the deployment requires incremental or federated model updates, where the compact trainable component offers communication and computation efficiency that tree ensembles are unlikely to match. The architecture is not proposed as a universal replacement for Random Forest on tabular IDS benchmarks, but rather as a structurally motivated choice for the sequential and distributed monitoring scenarios that increasingly characterise modern network security infrastructure.

Table 11 summarises these properties relative to the Random Forest baseline and representative prior work.

Figure 5 compares average reported accuracy across these paradigms, while Figure 6 traces the paradigm progression reviewed in Section 1.1 that motivates the proposed coupling.

Together, Figure 5 and Figure 6 summarise, respectively, where the proposed model sits relative to each paradigm’s best result and how the four paradigms reviewed in Section 1.1 led to the reservoir-to-attention coupling examined in this paper.

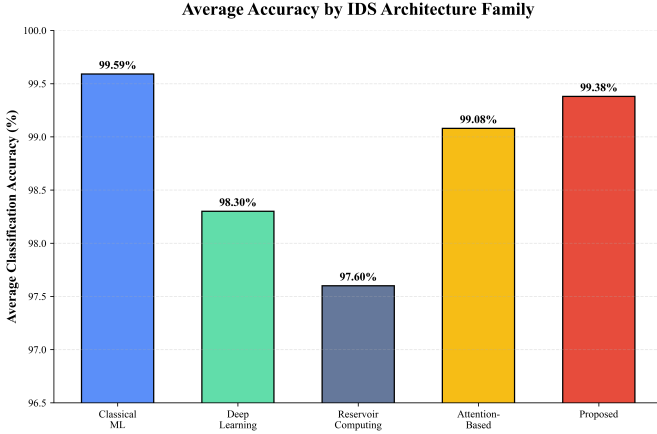


Figure 5. Average reported accuracy on CIC-IDS2017 for the best representative model within each of the four learning paradigms reviewed. Transformer-ESN (99.38%) is competitive with the strongest deep recurrent/convolutional baseline (CNN-BiGRU [9], 99.70%) and the strongest classical baseline (SVM-ELM [21], 99.67%), and outperforms the strongest standalone attention-based model (Transformer [10], 99.17%). As with Figure 1, cross-study comparisons are contextual rather than statistically controlled, and the proposed model remains the only entry with a complete per-class breakdown (Table 9).

Evolution of Intrusion Detection Architectures

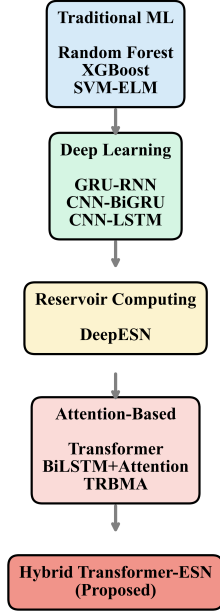


Figure 6. Timeline summarising the progression of network intrusion detection paradigms reviewed in Section 1.1: classical machine learning, deep recurrent/convolutional networks, reservoir computing, and attention-based architectures, culminating in the reservoir-to-attention coupling proposed in this work. The figure is intended as a conceptual overview of paradigm succession rather than a chronologically exhaustive record.

Workflow of the Proposed Transformer-ESN Intrusion Detection System

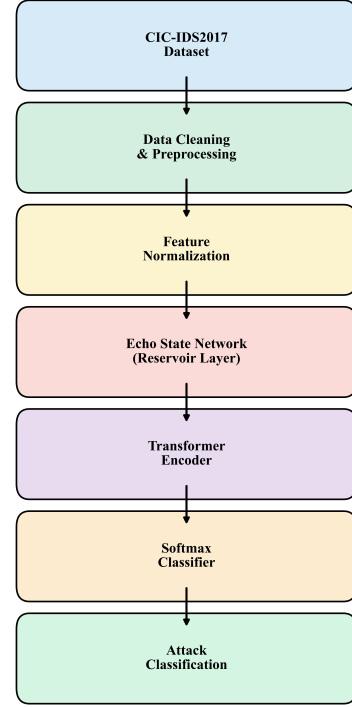


Figure 7. Training workflow of the proposed Transformer-ESN, from data loading through convergence checking and final evaluation. Reservoir weights W_{in} and W are initialised once and frozen throughout; only the Transformer encoder and classification head are updated via backpropagation at each iteration.

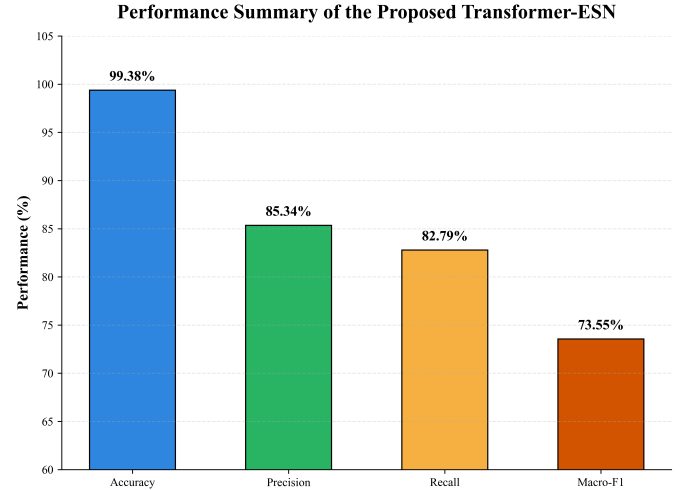


Figure 8. Per-class F1-score of the proposed Transformer-ESN on the CIC-IDS2017 test set, summarising the results of Table 9. Classes marked † have fewer than 393 test samples, so their F1 estimates are statistically unstable. The bimodal pattern—F1 ≥ 0.94 on nine of ten well-supported classes against near-zero F1 on four ultra-rare classes—accounts for the divergence between overall accuracy (99.38%) and macro-F1 (73.55%).

Table 11. Qualitative comparison of architectural properties. Checkmarks denote property presence; the proposed Transformer-ESN is the only entry with all four properties simultaneously.

Architecture	Temporal encoding	Gradient-free reservoir	Attention mechanism	Per-class eval.
Random Forest	✗	✗	✗	✗
GRU-RNN [7]	✓	✗	✗	✗
CNN-BiGRU [9]	✓	✗	✗	✗
DeepESN [27]	✓	✓	✗	✗
Transformer [10]	✗	✗	✓	✗
TRBMA [31]	✓	✗	✓	✗
Transformer-ESN (proposed)	✓	✓	✓	✓

4.6. Contextualisation Against Literature

Transformer-ESN’s 99.38% accuracy sits within the 96.21%–99.70% range reported across all four paradigms reviewed in Table 8. It outperforms the standalone Transformer of Wu et al. [10] and TRBMA [31] on accuracy, and is marginally below CNN-BiGRU [9] and the top classical results.

Direct numerical comparison with published figures should be treated with caution given protocol heterogeneity across studies [17]. What can be reasonably stated is that the proposed architecture’s accuracy does not appear degraded relative to the published range, and that its per-class performance on well-represented classes is competitive with the strongest published results. The reported limitation is specifically on ultra-rare classes—a limitation that is plausibly shared by many published architectures, though rarely reported, since most published evaluations omit per-class analysis.

Taken together, these results point to a broader conclusion about model selection in network intrusion detection. The macro-F1 gap observed in this study appears to be largely explained by two identifiable and partially separable factors: the statistical instability of ultra-rare class estimates, and a structural mismatch between sequential inductive biases and the pre-aggregated, non-sequential nature of CICFlowMeter features. Random Forest’s advantage on this benchmark is best understood as a consequence of its structural fit to tabular, row-independent data, rather than a general superiority in representational capacity. Conversely, Transformer-ESN’s design remains well motivated for data representations where genuine sequential structure exists, such as packet-level captures or session-ordered logs, and its training stability and parameter efficiency offer practical benefits in resource-constrained or federated settings regardless of the tabular benchmark result. The broader implication is that architecture selection in intrusion detection should be guided primarily by whether the available data representation carries genuine sequential information, rather than by benchmark accuracy alone; this framing motivates the directions for future work outlined below.

4.7. Implications for Future Work

Three actionable directions follow from these findings.

Packet-level sequential data. The sequential inductive bias of the Transformer-ESN architecture is expected to be most

beneficial when applied to data with genuine sequential structure. Evaluation on packet-level sequences within flows, or on temporally-ordered multi-flow session logs, would provide the architectural preconditions for the ESN reservoir and Transformer encoder to contribute as intended. This expectation is consistent with recent packet-level frameworks showing that flow-independent sequential processing improves detection suitability relative to purely flow-level aggregation [43].

Few-shot strategies for ultra-rare classes. The extreme rarity of several CIC-IDS2017 classes (4–36 total samples) makes reliable evaluation difficult under standard train-test splits. Future work should combine the proposed architecture with explicit few-shot learning, data augmentation (e.g., flow-level WGAN-GP synthesis [17]), or open-set detection strategies for ultra-rare classes, evaluated under protocols designed for low-support regimes.

Federated and incremental deployment. The fixed gradient-free reservoir creates a natural basis for federated learning: only the compact trainable component (508K parameters) would need to be updated or aggregated across distributed nodes, while reservoir weights remain locally fixed. Fully trainable recurrent architectures are unlikely to offer this structural advantage. Investigating this property for distributed or edge-deployed IDS is a direct extension of this work. This aligns with the broader federated-IDS literature, which identifies communication efficiency and reduced trainable footprint as central motivations for federated adoption [44]. A recent federated deployment tested directly on CIC-IDS2017 among other benchmarks reported its lowest accuracy specifically on CIC-IDS2017 (98.37%, versus 99.65% on a vehicular network dataset), underscoring that federated compactness does not by itself guarantee strong performance on this benchmark [45].

5. Conclusion

Network intrusion detection remains one of the most demanding classification problems in applied machine learning: class distributions are severely skewed, minority-class attacks carry disproportionate operational risk, and deployment environments impose hard constraints on training latency and labelled data availability. Existing deep learning approaches address these challenges either through purely recurrent architectures—which couple temporal feature extraction and discriminative classification into a single gradient-dependent pipeline subject to vanishing gradient instability and high sensitivity to initialisation—or through Transformer-only models applied directly to static feature vectors, which lack an explicit mechanism for capturing the nonlinear temporal dynamics present in ordered network traffic.

This paper addressed both limitations through a hybrid Transformer–Echo State Network (Transformer-ESN) architecture in which the two concerns are deliberately separated. A fixed, randomly initialised ESN reservoir first maps the 76-dimensional preprocessed network flow feature space into a 256-dimensional nonlinear state representation without any gradient-based optimisation [23, 24]. The resulting reservoir

states are then processed by a multi-head self-attention Transformer encoder [29], which learns long-range inter-feature dependencies across the projected space before a softmax classification head produces per-class probability scores. While prior work has applied Echo State Networks to intrusion detection in standalone configurations [25, 26, 28] and Transformer encoders have been widely adopted for network traffic classification [10, 11, 12], the specific coupling proposed here—using a fixed ESN reservoir as a zero-cost nonlinear projection layer that feeds directly into a Transformer encoder for multi-class network intrusion detection—has not, to the best of our knowledge, been previously reported in the literature.

Evaluated on the CIC-IDS2017 benchmark [13], the Transformer-ESN architecture achieved 99.38% overall accuracy, competitive with the range of published results on this dataset [18, 19, 21, 7, 8, 9, 10, 31]. The total trainable footprint is approximately 508K parameters, with the remainder of the model consisting entirely of fixed reservoir weights that require no gradient computation at any point during training or inference. Against a Random Forest baseline [18] evaluated under identical experimental conditions, the macro-F1 gap (73.55% vs. 88.40%) is explained by two specific and tractable factors: statistical unreliability of per-class metrics at ultra-low test support (4–301 samples for four classes), and architectural mismatch between sequential inductive biases and row-wise tabular features that lack genuine sequential order. Across the ten well-represented classes, the architecture achieves F1-scores of 0.61–1.00, with nine of ten exceeding 0.94, confirming that the performance gap is localised entirely to the sparse-support regime rather than reflecting a generalised architectural shortcoming.

Beyond raw accuracy, the architecture’s most practically relevant property is the clean separation between its two stages. Because reservoir weights are fixed at initialisation [23, 35], the model eliminates recurrent backpropagation entirely, reducing training complexity and improving convergence stability relative to end-to-end recurrent baselines [7, 5]. Only the Transformer encoder and classification head carry trainable parameters, making the total trainable parameter count compact and well-suited to settings where labelled traffic samples are scarce and model retraining must be completed under strict time constraints. This separation also creates a principled basis for incremental and federated learning, since only the compact trainable component needs to be updated or aggregated across distributed nodes—an important practical advantage that purely end-to-end architectures do not share [1, 2].

The principal contribution of this work is not a performance claim but a structural one: the reservoir-to-attention coupling is a training-stable, parameter-efficient approach that eliminates recurrent backpropagation while preserving the capacity for global contextual reasoning [27, 29]. Its limitations on CIC-IDS2017 are specific and tractable—attributable to feature representation mismatch and dataset sparsity [16, 17] rather than to generalised architectural shortcomings. The architecture is most appropriately evaluated in settings where genuine sequential structure exists in the input, such as packet-level streams or temporally-ordered session logs, where its sequential inductive bias becomes an advantage rather than a liability.

Taken together, the experimental results and architectural analysis establish Transformer-ESN as a competitive, principled, and computationally efficient framework for multi-class network intrusion detection. The proposed reservoir-to-attention coupling represents a complementary direction to existing hybrid architectures [8, 9, 30] and opens a clear path toward deployable, updatable, and interpretable detection systems in operational cybersecurity environments.

5.1. Future Scope

5.1.1. Packet-Level and Sequential Data Evaluation

The structural mismatch identified in this work—between the architecture’s sequential inductive bias and the row-wise tabular nature of CIC-IDS2017 flow features—is expected to diminish when the model operates on genuinely ordered inputs. Future work should evaluate Transformer-ESN on packet-level streams and temporally-ordered session logs, where ESN reservoir dynamics and Transformer attention can exploit real temporal ordering rather than imposing it artificially on independent feature vectors [7, 5].

This expectation is supported, with an important qualification, by recent work reformulating CIC-IDS2017 as a temporal task under controlled padding and split conditions [46]. On genuinely sequential, non-padded input windows, a standalone Transformer achieved the highest macro-F1 of any evaluated architecture, consistent with the expectation that sequential inductive biases become an advantage once real temporal structure is present. However, the same study found that padding convention and split protocol—not architecture—were the dominant factors governing reported performance: naive zero-padding degraded the Transformer’s macro-F1 by 0.24, and false-alarm rate under a leakage-free split increased by a factor of 67 relative to a conventional random split. This indicates that any future packet-level or session-level evaluation of Transformer-ESN must explicitly control for padding scheme and enforce leakage-free splitting; absent these controls, reported gains from sequential modeling are not distinguishable from evaluation-protocol artifacts [46].

5.1.2. Rare-Class Robustness and Evaluation Protocols

Four attack categories in CIC-IDS2017 had test support of 4–301 samples, making per-class F1-scores in those categories statistically unreliable. Future work should combine few-shot learning strategies, targeted augmentation for ultra-rare attack classes, and evaluation protocols specifically designed for the low-support regime—such as stratified cross-validation with bootstrapped confidence intervals—to produce assessments that distinguish genuine architectural limitations from artefacts of dataset sparsity [16, 17, 47].

5.1.3. Statistical Rigour and Repeated Experimental Evaluation

All results in this paper represent a single experimental run under a fixed random seed. While this is consistent with a substantial portion of published IDS evaluations, it precludes reporting confidence intervals or testing for statistical significance of performance differences between architectures. Future work

should conduct repeated runs across a range of random seeds (recommended minimum: 10–30 runs) and report mean performance with standard deviation or bootstrapped 95% confidence intervals for each metric. For macro-F1 in particular, where minority-class outcomes can shift aggregate values substantially on single runs, statistical characterisation of variance is a necessary condition for drawing reliable conclusions about architectural comparisons. Significance testing (e.g., paired Wilcoxon signed-rank tests across cross-validation folds) should be applied when comparing Transformer-ESN against the Random Forest baseline to determine whether the observed macro-F1 gap is statistically reliable or within run-to-run variance bounds.

5.1.4. Federated and Privacy-Preserving IDS

Collaborative intrusion detection across organisational boundaries is constrained by data privacy regulations that prohibit raw traffic sharing. Future research should investigate federated Transformer-ESN frameworks in which multiple stakeholders jointly optimise the trainable encoder using privacy-preserving gradient aggregation. The compact parameterisation of the trainable component—limited to the Transformer encoder and classification head, with reservoir weights fixed and never requiring transmission—is particularly advantageous for communication-efficient federated aggregation [1, 2].

5.1.5. Adaptive and Structured Reservoir Design

The random sparse reservoir topology adopted in this work may not optimally represent all classes of network traffic dynamics. Future work should systematically evaluate structured reservoir topologies—including ring networks, small-world graphs, and spectral graph-designed reservoirs—alongside adaptive initialisation strategies that tune spectral radius and connectivity density to the statistical properties of the target traffic distribution [48, 27, 35]. Adaptive reservoir designs may prove particularly effective for traffic distributions with non-stationary temporal structure, where fixed random projections are a limiting assumption.

5.1.6. Explainable Intrusion Detection

Operational security analysts require human-interpretable justifications for automated classification decisions before acting on model outputs. Future work should integrate attention heatmap visualisation and feature attribution methods to identify which reservoir states—and by extension, which temporal traffic patterns—are most predictive of specific attack categories. The multi-head attention weight matrices produced by the Transformer encoder offer a natural and already-available interpretability mechanism that can be leveraged without adding architectural overhead [10, 30, 49].

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